BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Variables, Input, Output and One-Dimensional Arrays

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before turning to the answers.

1. What does bubble sort compare?

2. What does insertion sort build?

3. What is the first array index in OCR ERL?

4. Trace one bubble-sort pass through [3, 2, 1].

Today's objectives

Assign and update variables using OCR ERL. Identify input and output in short programs.

Key vocabulary - Read, cover, recall

variable
Named storage for a value in a program.

assignment
Storing or updating a variable's value.

one-dimensional array
Indexed collection of values stored under one name.

index
Number specifying an array element's position.

Write one definition in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Four key ideas

VARIABLE	A named value can change using assignment.
INPUT / OUTPUT	Read data from a user; display or send the result.
1D ARRAY	Related values stored at numbered indexes.
INDEXES	The first element has index 0, not index 1.

ARRAY ELEMENTS - INDEX AND VALUE ARE DIFFERENT

Index 0	Index 1	Index 2
Value 4	Value 7	Value 9

total = marks[0] + marks[1] = 4 + 7 = 11

I Do - Worked example

Trace the array and variable assignments in a short quiz-mark program.

1. Start: marks = [4, 7, 9]. Index 0 contains 4.
2. Read marks[0] + marks[1] = 4 + 7 = 11.
3. Change marks[0] to 5; indexes stay the same.
4. Recalculate: 5 + 7 = 12. Output 12.

We Do - Three guided questions

1. Write an assignment that stores 12 in a variable called score.

2. For array marks = [6, 3, 8], identify marks[1].

3. Update marks[0] to 10 and calculate marks[0] + marks[2].

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the total of two file sizes stored in a one-dimensional array.

```
array fileMB = [2,5,8]
totalMB = fileMB[0] + fileMB[1]
print(totalMB)
```

1. Predict the output in MB.

2. Change fileMB[1] to 6.

3. Convert the revised total into KB using 1 MB = 1000 KB.

You Do - Independent exam questions

Q2. State one difference between a variable and an array. [2 marks]

Q3. Given values = [7, 2, 5], state the values at indices 0 and 2. [2 marks]

Q4. Write OCR ERL assignments to set total to 0, add values[0] and values[1], then display total. [4 marks]

Exit Ticket - True or False?

Answer before checking Page 4.

1. The element at index 2 is normally the third element of an OCR ERL array. ☐ T ☐ F

2. Index 0 is the first array item. ☐ T ☐ F

3. Arrays and variables always store exactly the same number of values. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Mark fairly. Tick correct points and fix any mistakes.
Use the mark allocations; then rewrite one answer in full.

Do Now - Answers

Q1 Neighbouring values. ☐

Q2 A growing sorted section. ☐

Q3 Zero. ☐

Q4 [2, 3, 1] then [2, 1, 3]. ☐

Code First - Model

1. Initial output: 7 MB (2 + 5). ☐

2. After change: 8 MB (2 + 6). ☐

3. 8 MB = 8,000 KB at 1000 KB per MB. ☐

We Do - Three model answers

1. score = 12.

2. 3.

3. 10 + 8 = 18.

Exit Ticket - Check

1. TRUE 2. TRUE 3. FALSE

Explain one false statement using the key ideas.

You Do - Self-mark

Q2 Variable stores a value [1]; array stores indexed values [1]. ☐

Q3 Index 0 = 7 [1]; index 2 = 5 [1]. ☐

Q4 total=0 [1]; add values[0] [1]; add values[1] [1]; print(total) [1]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW answer. Include working if needed: